

EXPERIMENT: 35

IMPLEMENTATION OF SLIDING WINDOW PROTOCOL IN JAVA/C

```
#include <stdio.h>

int main()
{
    int frames[20];
    int n, window;
    int i, start;

    printf("SLIDING WINDOW PROTOCOL\n");
    printf("-----\n");

    printf("Enter number of frames: ");
    scanf("%d", &n);

    printf("Enter window size: ");
    scanf("%d", &window);

    printf("Enter frame numbers:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &frames[i]);
    }

    printf("\nTransmission:\n");
```

```
start = 0;

while(start < n)
{
    printf("\nWindow: ");

    for(i = start; i < start + window && i < n; i++)
    {
        printf("%d ", frames[i]);
    }

    printf("\n");

    for(i = start; i < start + window && i < n; i++)
    {
        printf("Sending Frame %d\n", frames[i]);
    }

    printf("ACK received for frames ");

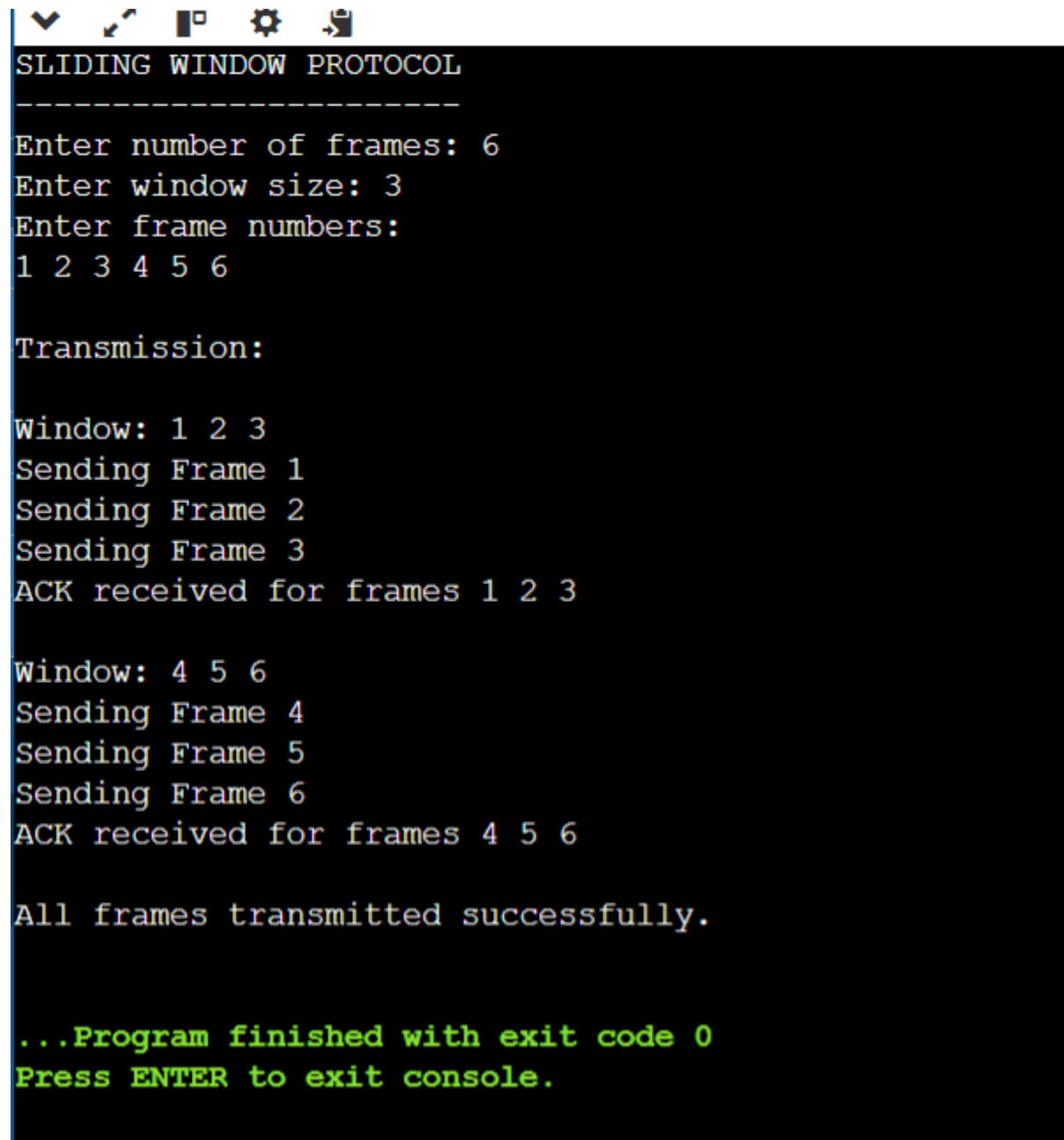
    for(i = start; i < start + window && i < n; i++)
    {
        printf("%d ", frames[i]);
    }

    printf("\n");

    start = start + window;
}
```

```
printf("\nAll frames transmitted successfully.\n");

return 0;
}
```

A screenshot of a terminal window with a dark background and light-colored text. The window title bar at the top shows standard icons for window control (minimize, maximize, close) and a gear icon for settings. The text inside the terminal is as follows:

```
SLIDING WINDOW PROTOCOL
-----
Enter number of frames: 6
Enter window size: 3
Enter frame numbers:
1 2 3 4 5 6

Transmission:

Window: 1 2 3
Sending Frame 1
Sending Frame 2
Sending Frame 3
ACK received for frames 1 2 3

Window: 4 5 6
Sending Frame 4
Sending Frame 5
Sending Frame 6
ACK received for frames 4 5 6

All frames transmitted successfully.

...Program finished with exit code 0
Press ENTER to exit console.
```